



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

Prof Masoumeh Kazemi



Agenda

- Chapter 8 Quality Control

Quality

Quality is the totality of features and characteristics of a product that bear on its ability to satisfy stated or implied needs.

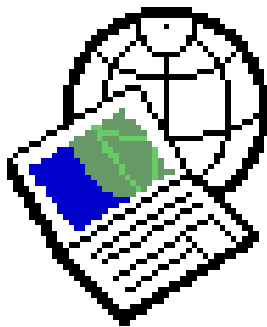
International Standards Organization (ISO)



Quality

Quality...is the extent to which the customer or users believe the product or service surpasses their needs and expectations.

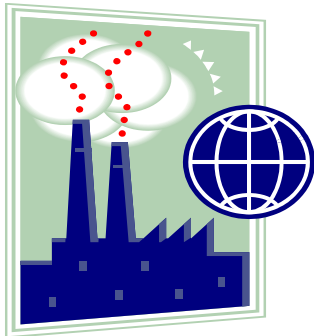
Gitlow *et. al.*



Quality

Quality is the total composite product and service characteristics of marketing, engineering, manufacture and maintenance through which the product in use will meet the expectations of the customer.

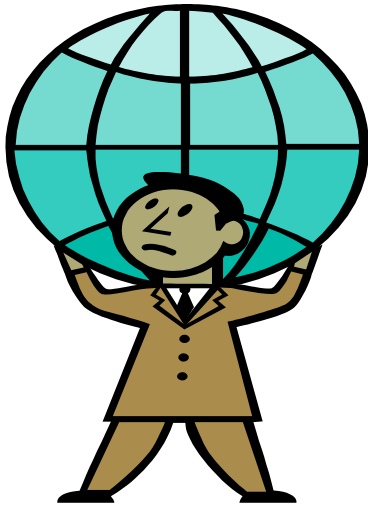
Feigenbaum



Quality

Quality is anything which can be improved.

Imai



Quality

Quality is conformance to requirement.

Crosby



Quality

Quality ... a thing is said to have the positive attribute of conformance to specified standards.

Shewhart



Quality

Quality is fitness for use.

Juran



Quality

Good quality ... means a predictable degree of uniformity and dependability at a low cost with a quality suited to the market.

Deming (Father of the Quality Improvement)



Events that fostered quality

- Baldrige National Quality award (US)
 - Total Quality Management (TQM)
- Deming's thoughts on Continuous improvement (Dr. Edwards Deming)
 - 14-point management principles for total quality management (TQM)
- Juran's contributions to quality thoughts
 - Quality planning; quality control; quality improvement

The manufacturing-based approach



The manufacturing-based approach focuses on the supply side of the equation, and are primarily concerned with engineering and manufacturing practice.



The value-based approach



Quality is defined in terms of costs and prices. A quality product is one that provides performance at an acceptable price or conformance at an acceptable cost.

Dimensions of Quality for Goods

- Performance
- Reliability & durability
- Conformance
- Serviceability
- Appearance
- Features
- Safety

Service Quality Attributes

Reliability

Competence

Responsiveness

Courtesy

Access

Communication

Reliability

- Reflects the probability of a product's failing within a specified period of time (Mean Time to First Failure, MTFF and Mean Time between Failures, MTBF) – Japanese manufacturers pay great attention to this dimension.
- Relevant to durable goods than they are to products and services that are consumed instantly.

Conformance

- Represents the degree to which a product's design and operating characteristics match pre-established specifications and standards.
- Commonly measured by the incidence of defects: the proportion of all units that fail to meet specification, and so require repair or rework.

Durability

- Represents the amount of use a consumer gets from a product before it physically deteriorates (light bulb provides the perfect example).
- It is a measure of product life, and has both economic and technical dimensions.

Serviceability

- This dimension of quality is synonymous of speed, courtesy, and competence of repair.
- Some variables associated with this dimension can be measured quite objectively, others reflect differing personal standards of what constitutes acceptable service.

Aesthetics



- Aesthetics – how a product looks, feels, sounds, tastes, or smells – is clearly a matter of personal judgment and reflects individual preferences. It is thus highly **subjective**.
- It is closely related to the user-based approach.

Perceived quality

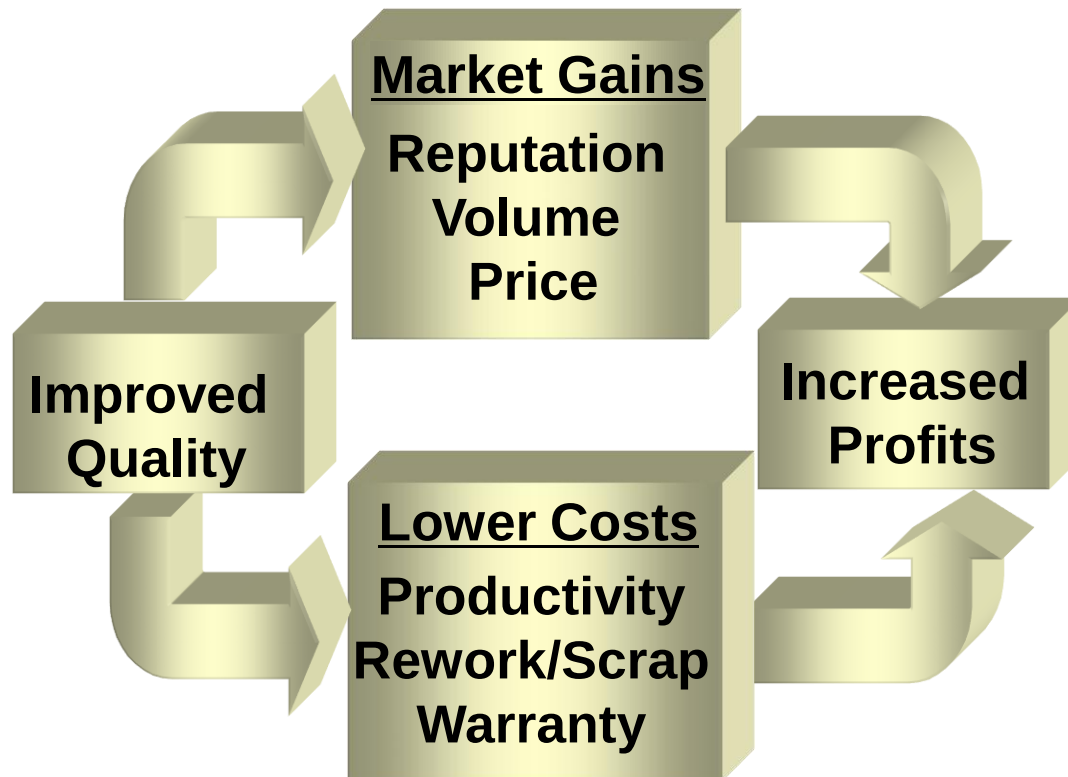


- Consumers frequently rely on indirect measures when comparing brands, because they do not always possess complete information about a product's attributes.
- Products are evaluated less on their objective characteristics than on their images, advertising, or brand names.

Determinants

- Quality of design
- Quality of capability of production process
- Quality of conformance
- Quality of customer service

Importance of Quality



Quality Benefit

- Published data support that by focusing on quality, a business can substantially increase its market share and profitability.

International Standards Organization - ISO

- Quality management system (QMS) standards
- Guidelines for quality (documented procedure)
- Specify *what* is required but not *how* to do it
- Certification is administered by a third party
- Must be renewed every three years
- **ISO 9001** → quality management (making sure products/services meet customer needs)
- **ISO 14001** → environmental management (reducing pollution and waste)
- **ISO 45001** → health and safety at work
- **ISO 27001** → information security (protecting data)

Related Concepts

- **Quality Management:** must be concerned with different factors such as determining what the customers' requirements are, setting product specifications to meet those requirements, developing processes that can produce the product with those specifications for an acceptable cost, purchasing material from suppliers at an acceptable cost, implement procedures to ensure that quality is monitored and maintained.

Quality Control and Assurance

- **Quality Control:** includes methods and procedures for testing and maintaining the level of quality that is desired
 - Focus: Detection of defects
 - Goal: Identify and fix issues in the final product or output
 - Activities: inspection, testing, sampling, rework
- **Quality Assurance:** is much broader and includes any activity related to making sure that quality is maintained at the desired level
 - Focus: Prevention of defects
 - Goal: Ensure that the process used to make the product is effective and consistent.
 - Activities: developing standard procedures, process audits, employee training, documentation, certification

Costs of Quality

- **Prevention costs** - reducing the potential for defects
- **Appraisal costs** - evaluating products (inspection/quality control)
- **Internal failure costs** - producing defective parts or service
- **External failure costs** - occur after delivery
- **Total quality costs**

Prevention costs

- Quality planning: costs associated with time to spend planning quality, data collection, tests procedures, etc.
- Quality training
- Design of quality systems: studying and analyzing production systems
- Quality reporting

Appraisal costs

- Testing and inspection
- Quality Audits: costs associated with measuring the level of quality and evaluating the systems and procedures used to monitor and maintain that quality.

Internal Failure Costs

- Scrap, rework, retest, downtime, yield losses (i.e. cost of material caused by faulty processes such as overfilling product containers)
- Disposition: costs involved in determining whether defects can be corrected or must be scrapped.

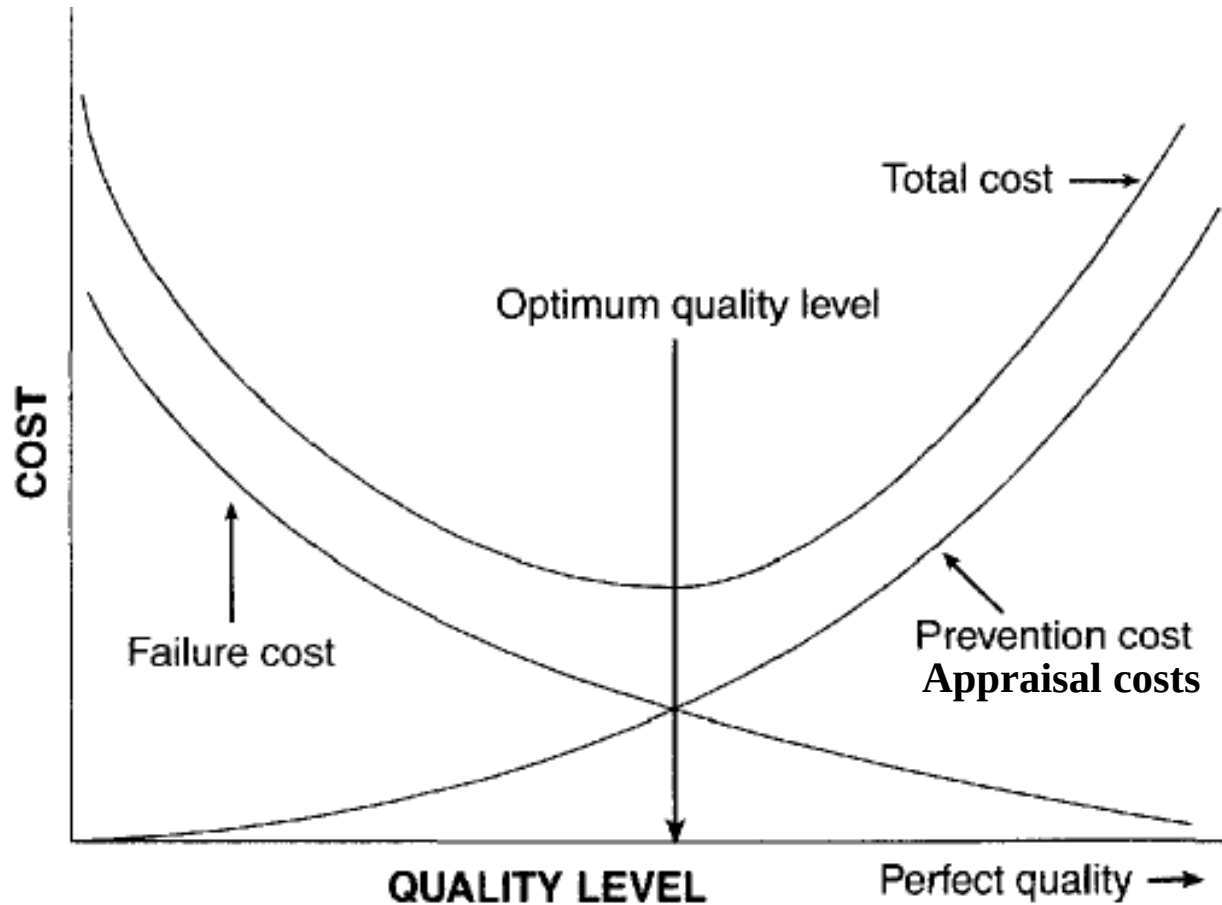
External Failure costs

- Complaint adjustment: costs of monitoring and responding to complaints
- Returned material
- Warranty charges
- Allowances: cost of providing an allowance to the customer if that product fails within a stated period
- Loss of goodwill

Total quality costs

It represents the difference between the actual cost of a product or service, and what the reduced cost would be if there was no possibility of substandard service, failure of products, or defects in their manufacture.

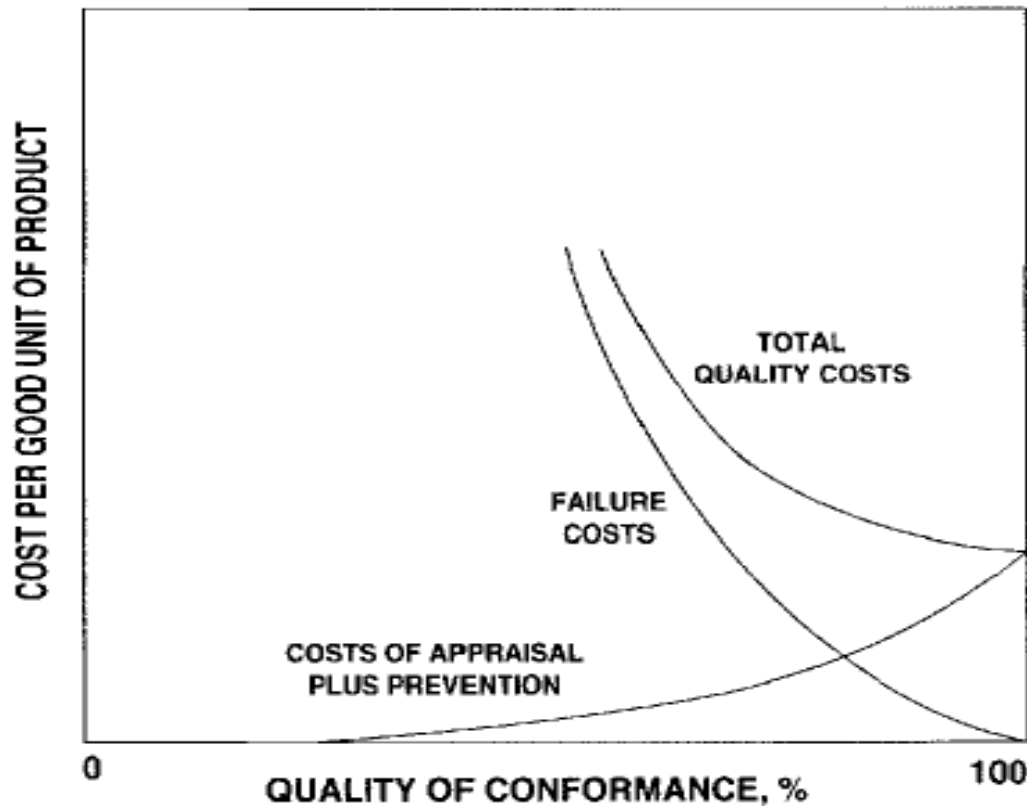
Classical model of optimum quality costs



From *Juran's Quality Control Handbook*, 4th edition. J.M. Juran, editor.

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New model of optimum quality costs



- Many organizations have demonstrated that with good design, processes and automation, you can drive defects to extremely low levels at manageable cost.

From *Juran's Quality Control Handbook, 4th edition*. J.M. Juran, editor.

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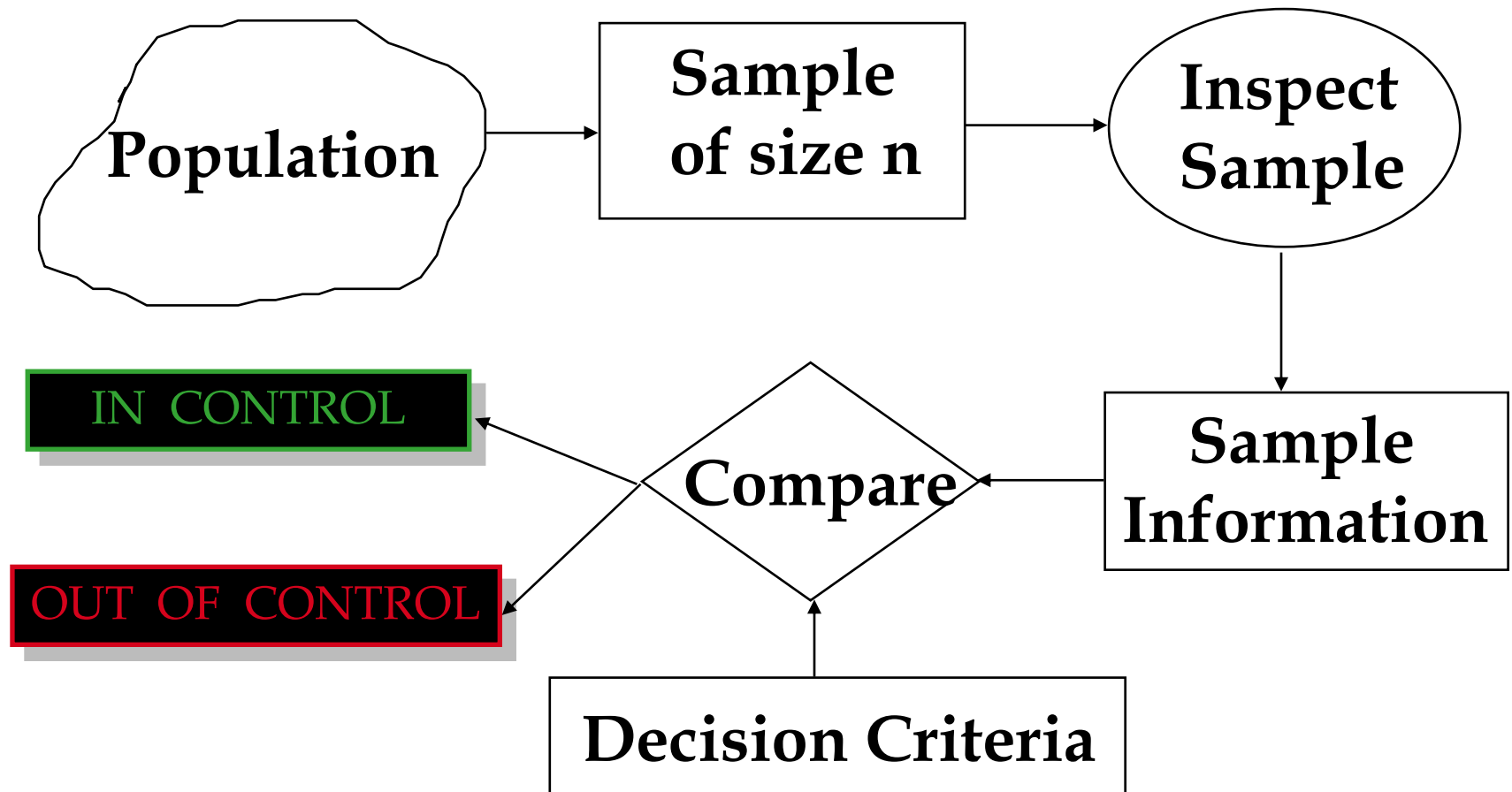
Quality Data and Analysis

- Quality cost data is designed to justify and support quality performance improvement
- Quality cost data helps to identify problem areas and direct resources to these areas
- Statistical analysis can be used to correlate quality cost trends with other quality data to help direct attention to problem causes

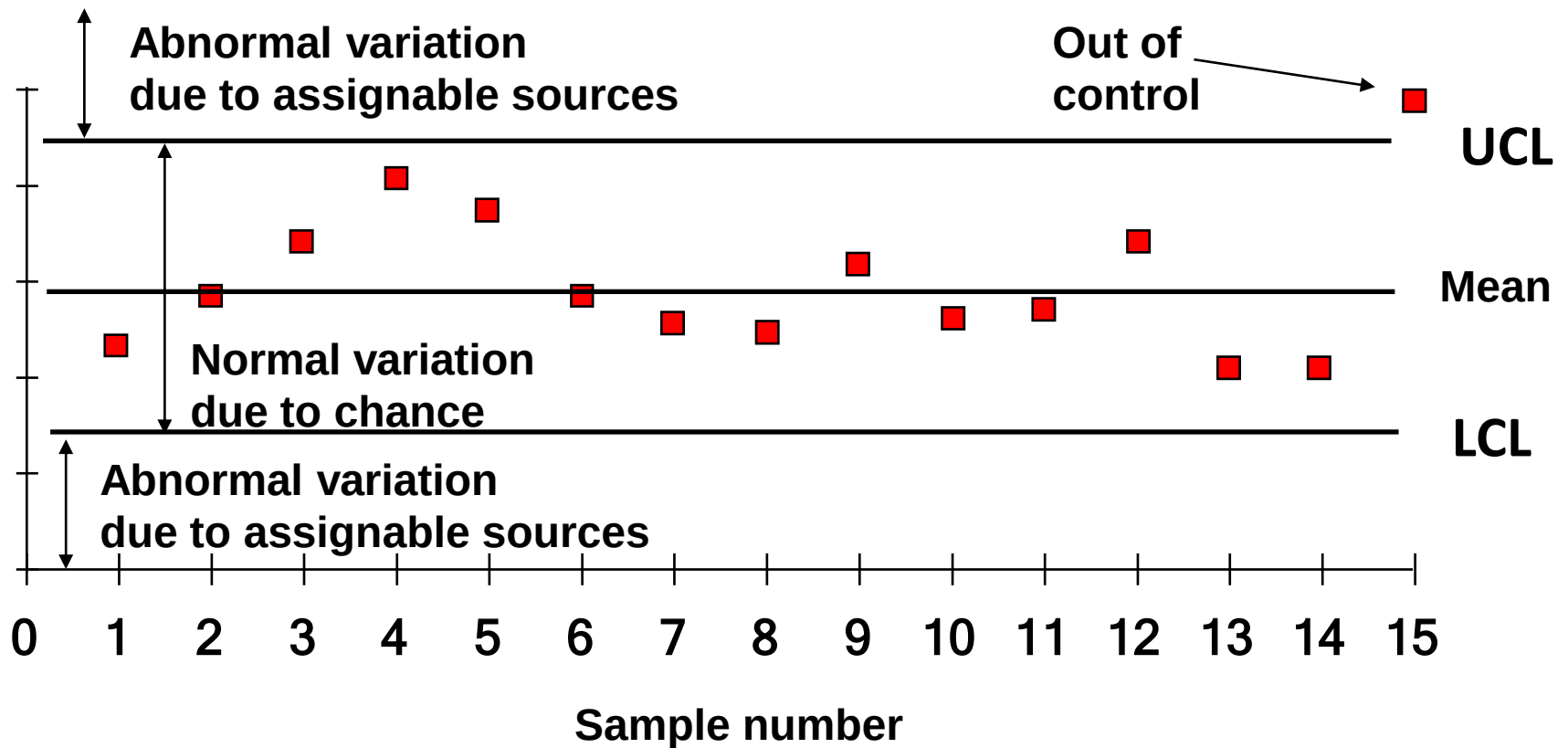
Statistical Process Control

- SPC is a data-driven method for distinguishing between natural process variation and assignable process variation
- Random variation: Natural (inherent) variations in the output of process, created by countless minor factors
 - Small fluctuations in raw material properties; gradual tool wear; temperature, humidity, air pressure; operator fatigue or reaction time; etc.
- Assignable variation: A variation whose source can be identified and has a significant impact on the process output
 - Equipment failure; lost calibration of machines; changeover mistake; new supplier; operator error; sudden power fluctuation; etc.

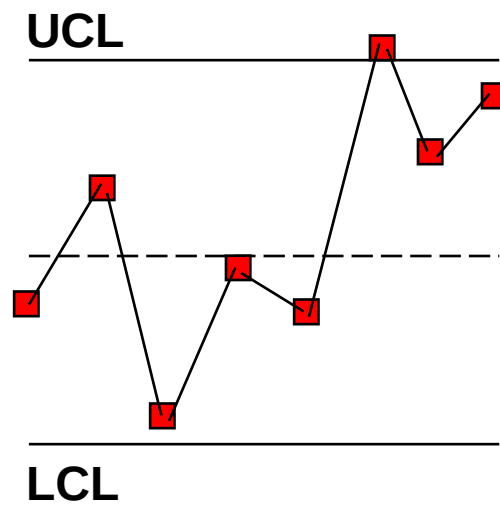
Overview of the Quality Control Process



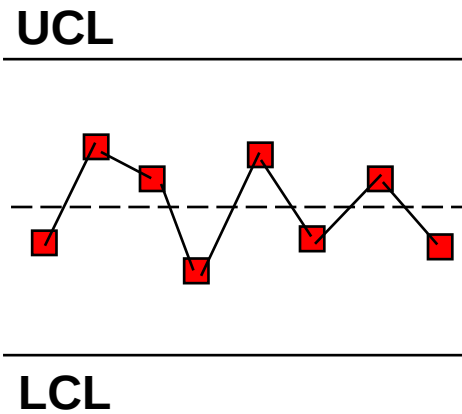
Control Chart



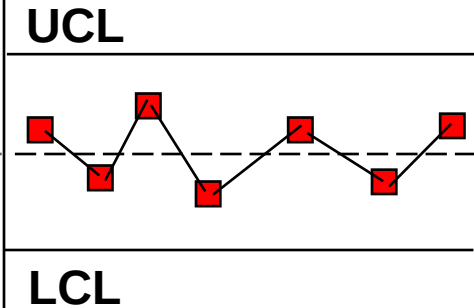
Tracking Improvements



Process not centered
and not stable



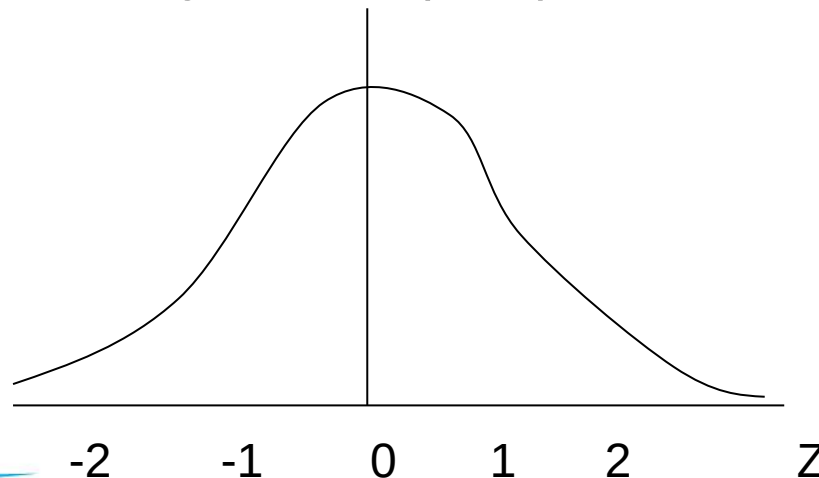
Process centered
and stable



Additional improvements
made to the process

When a population is normally distributed:

- The normal frequency curve shows that the highest frequency falls in the centre (i.e., at the mean of the values in the distribution) with an equal and exactly similar curve on either side of that centre. Thus, the most frequent value in a normal distribution is the average, with half the values falling below the average and half above it.
- The Normal curve, often called a bell curve, is perfectly symmetrical. Therefore the mean (the arithmetical average), the mode (the most frequent value), and the median (the middle value) will coincide at the centre of the curve (the high point).
- The further away any particular value is from the average (above or below), the less frequent that value will be (i.e., the frequencies will diminish on either side of the high central point).



When a population is normally distributed:

- Because the two halves on either side of the centre are exactly symmetrical, the frequency of values above the mean will match exactly the frequencies of values below the mean, provided the distances between the values and mean are identical. Thus, the frequency of a value 3 units to the right of the mean will be identical to the frequency of the value 3 units to the left of the mean.
- Normal curves may have different shapes (i.e., tall and skinny, short and low, and so on). What will determine the overall shape of the symmetrical curve will the value of the mean and the standard deviation in the population

Review of Normal Distribution

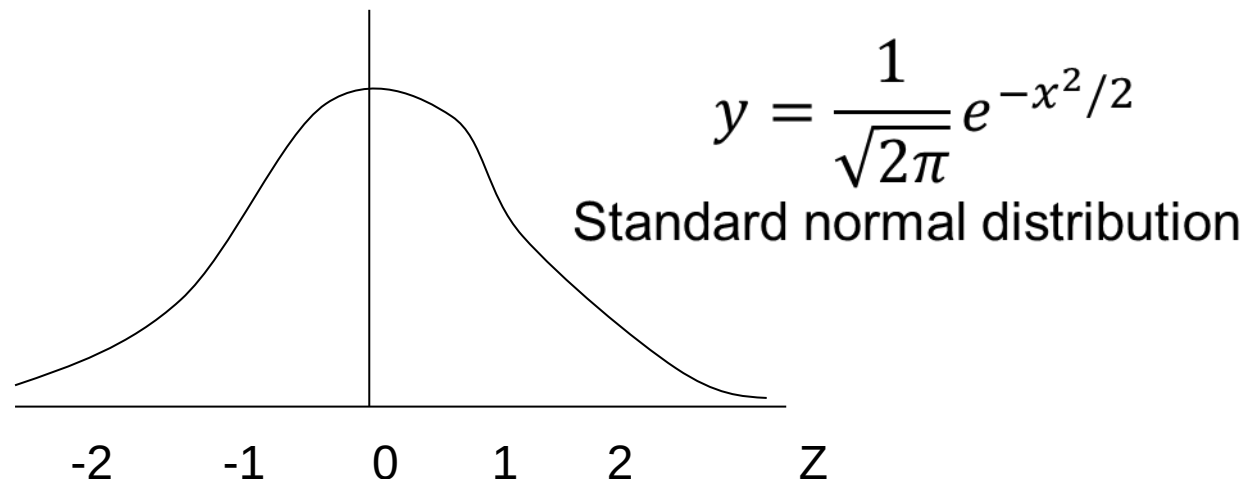
- Two parameters of the Normal distribution are the mean and variance (or standard deviation)
- The variance is a measure of how spread out a distribution is. It is computed as the average squared deviation of each number from the mean. For example, for the numbers 1, 2, and 3, the mean is 2 and the variance is:

$$\sigma^2 = [(1-2)^2 + (2-2)^2 + (3-2)^2] / 3 = 0.667$$

- The formula summation notation for the variance in a population is
- $\sigma^2 = \sum (X - \mu)^2 / n$
- σ : standard deviation

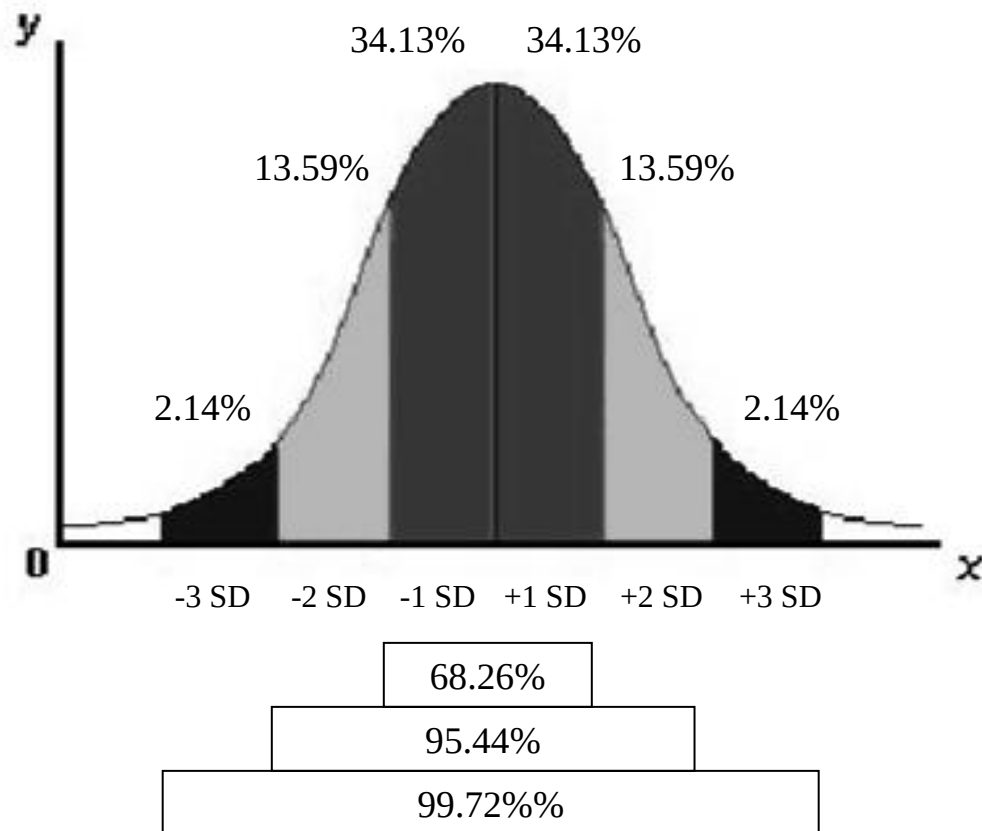
Review of Normal Distribution

- The standard Normal distribution is a Normal distribution with a mean of 0 and a standard deviation of 1. Normal distributions can be transformed to standard normal distributions by the formula:
$$Z = (X - \mu) / \sigma$$
- Z score always reflects the number of standard deviations above or below the mean a particular score is. For instance, if a person scored a 70 on a test with a mean of 50 and a standard deviation of 10, then they scored 2 standard deviations above the mean.



Review of Normal Distribution

The odds of a particular score, or set of scores, falling within a particular region are equal to the percentage of the total area occupied by that region



Different Types of Control Charts

- X-bar ($\bar{X}$) charts
- R charts
- p charts
- tendency around mean
- range of measurements
- pass/fail

Control Charts for variable data

- Measurements
- Inspection performed on variables basis (measure is continuous)
- Two control charts monitor two different aspects of the process' s output
 - **Central tendency**
 - monitored using the $\bar{X}$ -chart
 - **Variability**
 - monitored using the R-chart

SPC charts for variable data

Table 8.2. STRENGTH TEST DATA OF RAW MATERIAL

Subgroup	Sample Data				Subgroup Average, $\bar{X}$	Subgroup Range, R
	X_1	X_2	X_3	X_4		
1	908	1,095	1,159	1,215	1,094.25	307
2	1,337	1,080	1,086	1,242	1,186.25	257
3	1,212	1,326	1,175	1,380	1,273.25	205
4	1,286	1,278	1,151	1,359	1,268.50	208
5	1,401	1,313	1,384	1,466	1,391.00	153
6	1,114	1,096	1,060	1,204	1,118.50	144
7	1,119	974	1,007	1,212	1,078.00	238
8	1,001	1,030	1,063	1,037	1,032.75	62
9	1,152	1,161	1,159	1,046	1,129.50	115
10	1,000	882	870	823	893.75	177
11	835	667	787	803	773.00	168
12	707	904	961	918	872.50	254
13	949	809	842	859	864.75	140
14	828	1,061	918	964	942.75	233
15	1,043	753	813	861	867.50	290
16	784	847	856	945	858.00	161
17	1,086	925	855	840	926.50	246
18	1,117	1,201	1,034	929	1,070.25	272
19	1,115	1,114	1,038	1,227	1,123.50	189
20	1,140	928	1,332	1,061	1,115.25	404
21	1,229	1,127	1,030	1,253	1,159.75	223
22	1,314	1,242	1,087	1,140	1,195.75	227
23	1,305	1,219	1,353	1,207	1,271.00	146
24	1,181	1,326	1,245	1,548	1,325.00	367
25	1,312	1,285	1,444	1,176	1,304.25	268
26	993	1,052	1,018	1,074	1,034.25	81
27	1,196	980	1,116	1,161	1,113.25	216
28	948	1,139	1,222	1,062	1,092.75	274
29	993	1,283	1,231	1,105	1,153.00	290
30	1,050	928	1,009	1,058	1,011.25	130
					32,540.00	6,445.00

Control Charts for variable data

- $\bar{X}$ - Chart:
 - $3\bar{\sigma}$ upper and lower limits are $\bar{\bar{X}} \pm A_2\bar{R}$
 - R - Chart:
 - $3\bar{\sigma}$ lower control limit is $D_3\bar{R}$
 - $3\bar{\sigma}$ upper control limit is $D_4\bar{R}$
- A_2 , D_3 and D_4 are based on the sample size
- The standard deviation of the population can be estimated as follows:

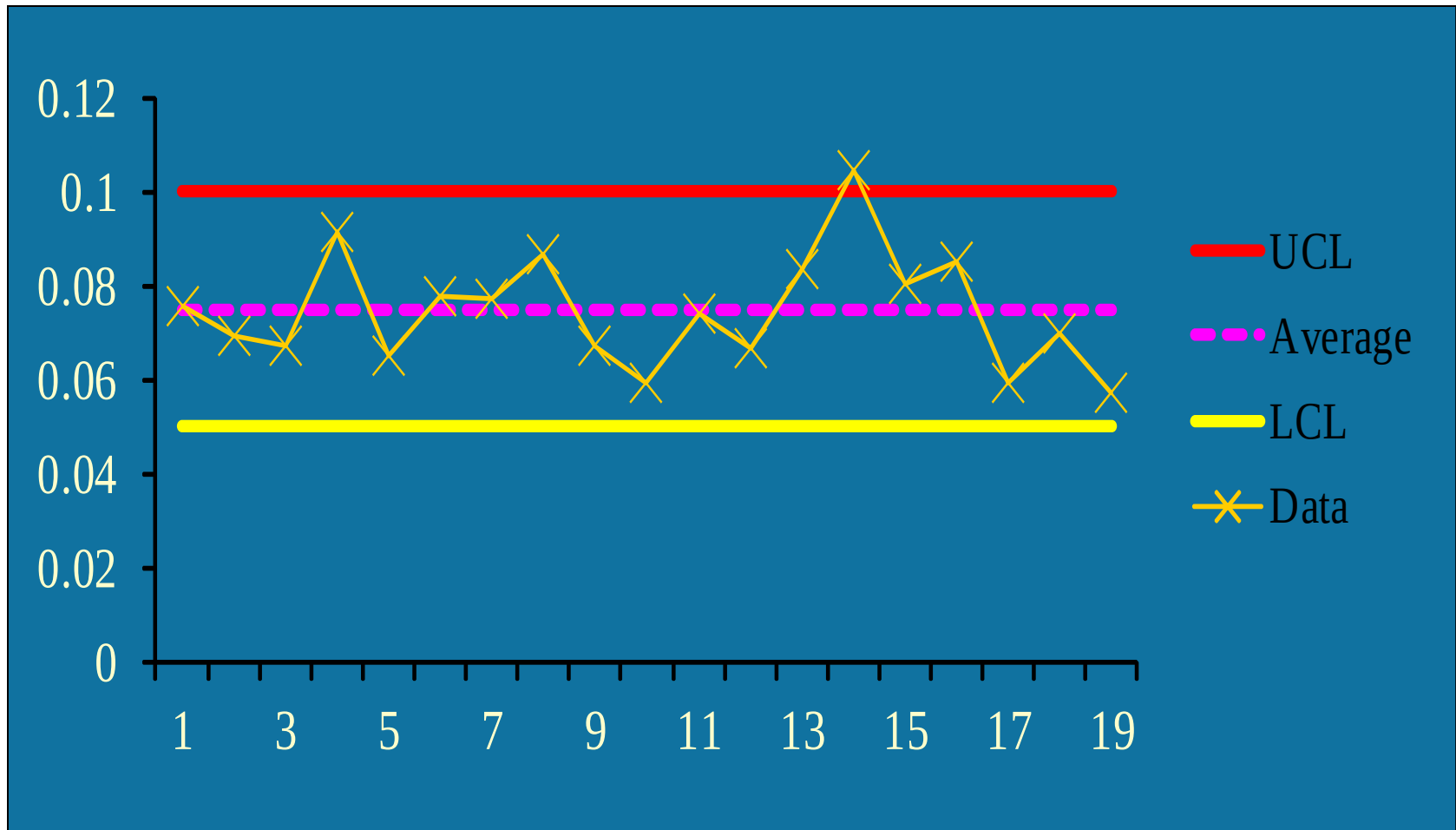
$$\hat{\sigma} = \frac{\bar{R}}{d_2}$$

SPC charts coefficients

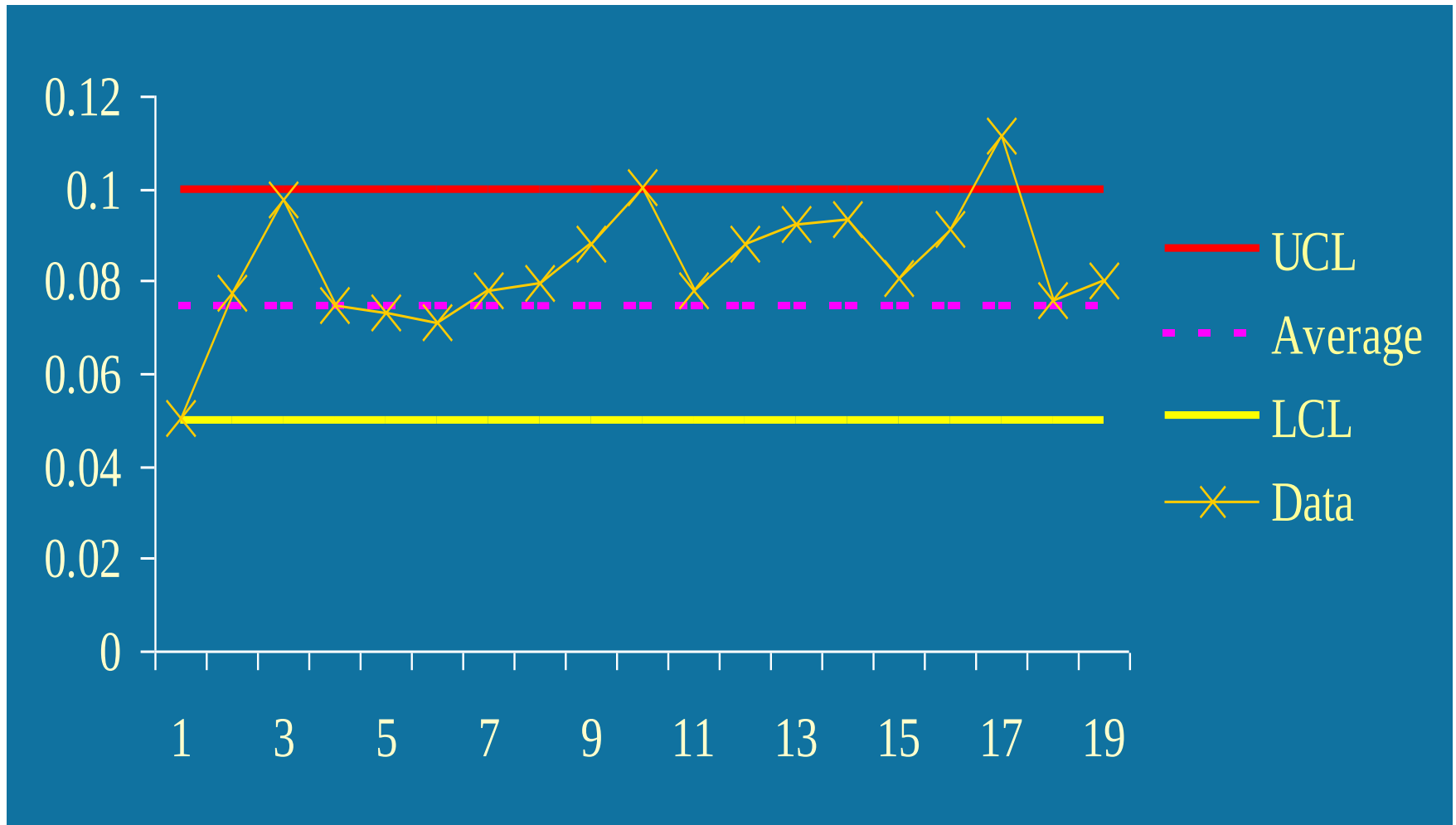
Table 8.1. CONTROL CHART FACTORS

Subgroup Size, n	LCL_R Factor, D_3	UCL_R Factor, D_4	$CL_{\bar{X}}$ Factor, A_2	$\bar{R}/\sigma$ Ratio, d_2
2	0	3.267	1.880	1.128
3	0	2.575	1.023	1.693
4	0	2.282	0.729	2.059
5	0	2.115	0.577	2.326
6	0	2.004	0.483	2.534
7	0.076	1.924	0.419	2.704
8	0.136	1.864	0.373	2.847
9	0.184	1.816	0.337	2.970
10	0.223	1.777	0.308	3.078
11	0.256	1.744	0.285	3.173
12	0.284	1.716	0.266	3.258
13	0.308	1.692	0.249	3.336
14	0.329	1.671	0.235	3.407
15	0.348	1.652	0.223	3.472
16	0.364	1.636	0.212	3.532

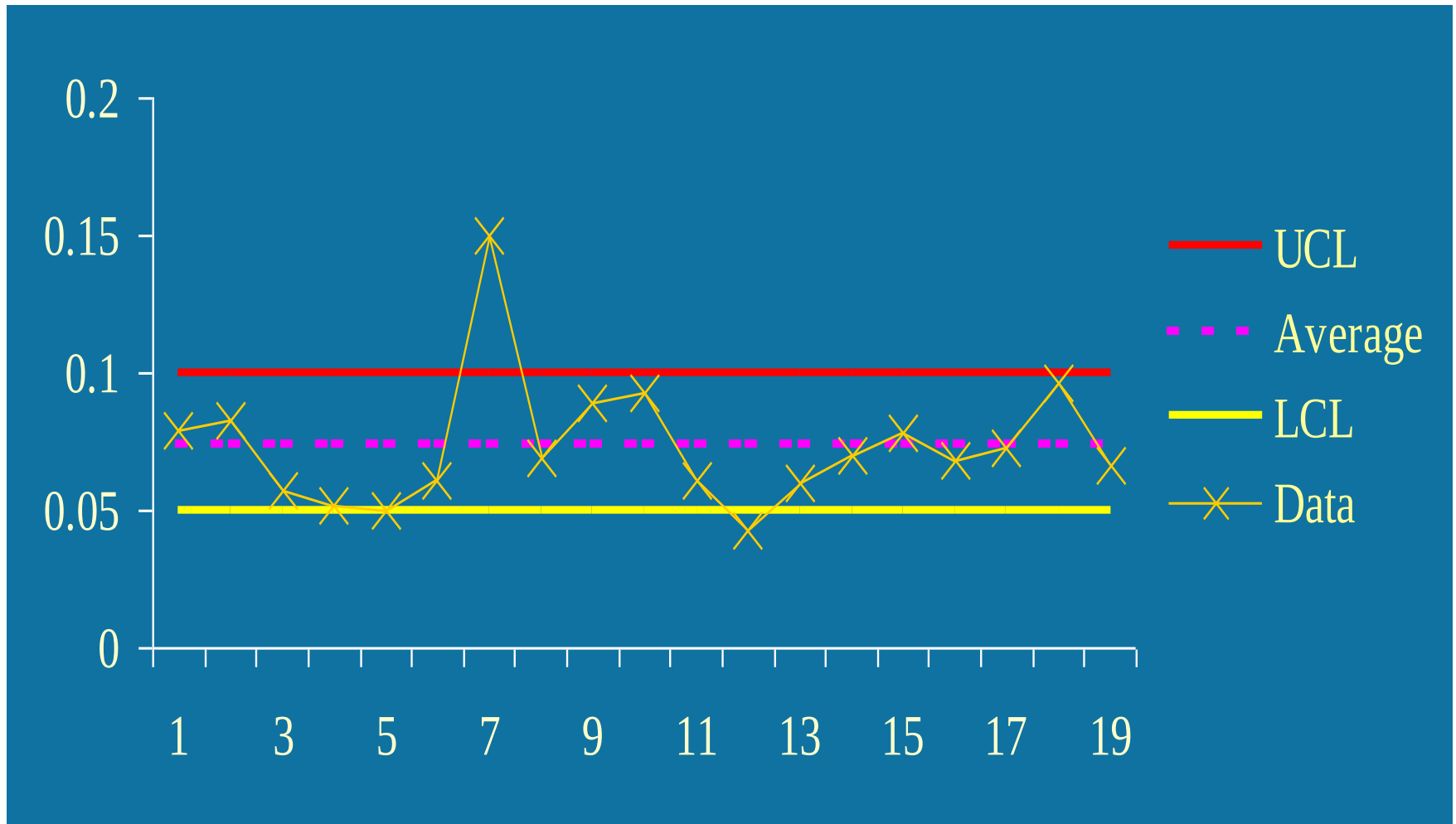
SPC Chart 1



SPC Chart 2



SPC Chart 3

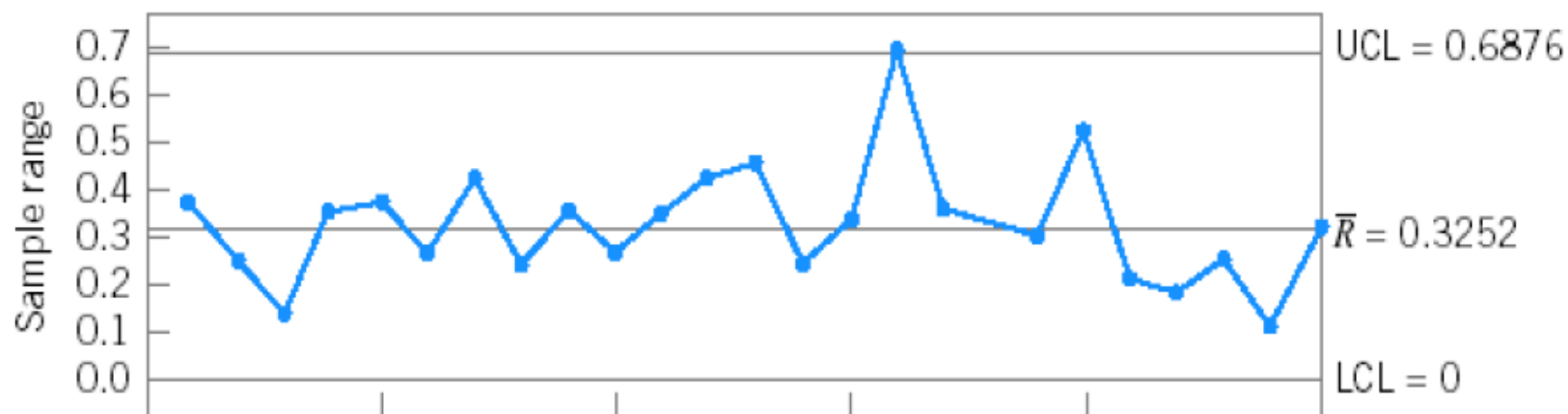
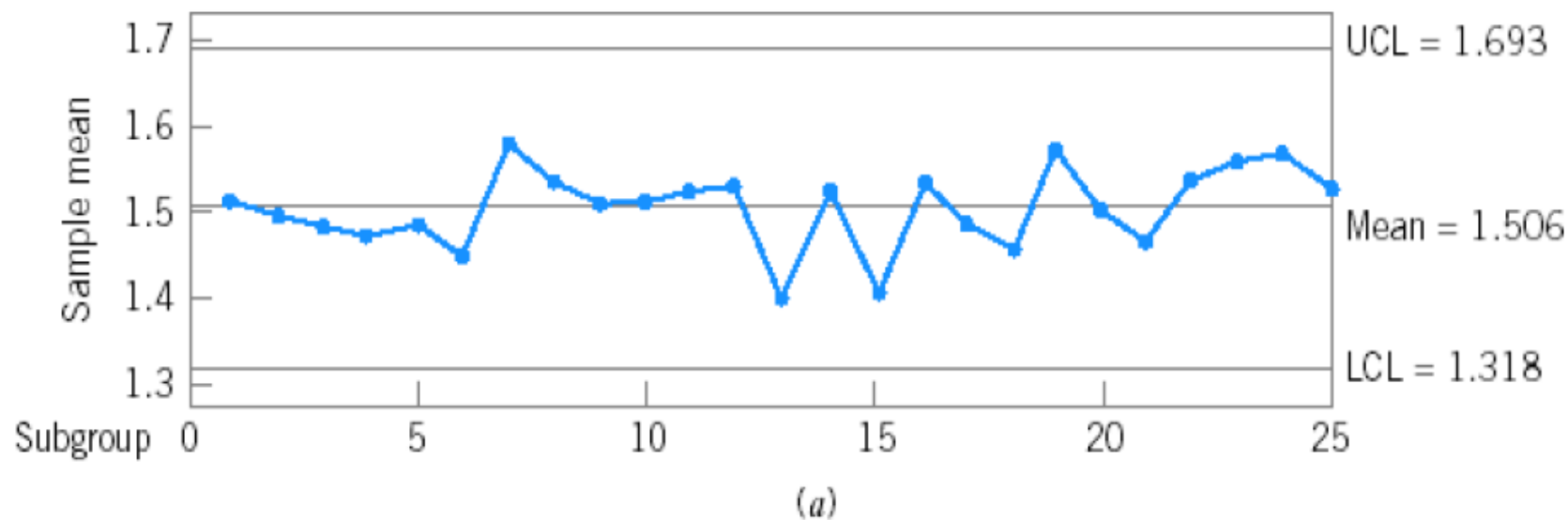


Example

Table 5-1 Flow Width Measurements (microns) for the Hard-Bake Process

Sample Number	Wafers					$\bar{x}_i$	R_i
	1	2	3	4	5		
1	1.3235	1.4128	1.6744	1.4573	1.6914	1.5119	0.3679
2	1.4314	1.3592	1.6075	1.4666	1.6109	1.4951	0.2517
3	1.4284	1.4871	1.4932	1.4324	1.5674	1.4817	0.1390
4	1.5028	1.6352	1.3841	1.2831	1.5507	1.4712	0.3521
5	1.5604	1.2735	1.5265	1.4363	1.6441	1.4882	0.3706
6	1.5955	1.5451	1.3574	1.3281	1.4198	1.4492	0.2674
7	1.6274	1.5064	1.8366	1.4177	1.5144	1.5805	0.4189
8	1.4190	1.4303	1.6637	1.6067	1.5519	1.5343	0.2447
9	1.3884	1.7277	1.5355	1.5176	1.3688	1.5076	0.3589
10	1.4039	1.6697	1.5089	1.4627	1.5220	1.5134	0.2658
11	1.4158	1.7667	1.4278	1.5928	1.4181	1.5242	0.3509
12	1.5821	1.3355	1.5777	1.3908	1.7559	1.5284	0.4204
13	1.2856	1.4106	1.4447	1.6398	1.1928	1.3947	0.4470
14	1.4951	1.4036	1.5893	1.6458	1.4969	1.5261	0.2422
15	1.3589	1.2863	1.5996	1.2497	1.5471	1.4083	0.3499
16	1.5747	1.5301	1.5171	1.1839	1.8662	1.5344	0.6823
17	1.3680	1.7269	1.3957	1.5014	1.4449	1.4874	0.3589
18	1.4163	1.3864	1.3057	1.6210	1.5573	1.4573	0.3153
19	1.5796	1.4185	1.6541	1.5116	1.7247	1.5777	0.3062
20	1.7106	1.4412	1.2361	1.3820	1.7601	1.5060	0.5240
21	1.4371	1.5051	1.3485	1.5670	1.4880	1.4691	0.2185
22	1.4738	1.5936	1.6583	1.4973	1.4720	1.5390	0.1863
23	1.5917	1.4333	1.5551	1.5295	1.6866	1.5592	0.2533
24	1.6399	1.5243	1.5705	1.5563	1.5530	1.5688	0.1156
25	1.5797	1.3663	1.6240	1.3732	1.6887	1.5264	0.3224
						$\Sigma \bar{x}_i = 37.6400$	$\Sigma R_i = 8.1302$
						$\bar{\bar{x}} = 1.5056$	$\bar{R} = 0.32521$

$\bar{x}$ and R charts



Example

- A company bottles soft drinks. The bottles come in only one flavor and only one size (16 ounces). The first daily samples of fill weights of 20 bottles are given in the table. Compute the 3-sigma limits for sample means and sample ranges. What would you conclude about the fill weights?

Sample	X-bar	R
1	16.05	0.20
2	16.04	0.25
3	15.98	0.62
4	15.91	0.71
5	16.02	0.58
6	16.09	0.37
7	15.95	0.35
8	16.06	0.21
9	15.94	0.29
10	15.97	0.46

Solution – sample means

- $n = 20$
- $A_2 = 0.18$
- $D_3 = 0.41$
- $D_4 = 1.59$

$\bar{R}$

$= 0.4$

$\bar{\bar{X}} = 16$

Sample	X-bar	R
1	16.05	0.20
2	16.04	0.25
3	15.98	0.62
4	15.91	0.71
5	16.02	0.58
6	16.09	0.37
7	15.95	0.35
8	16.06	0.21
9	15.94	0.29
10	15.97	0.46

- $LCL_R = D_3 \bar{R} = 0.41(0.4) = 0.164$ ounces
- $UCL_R = D_4 \bar{R} = 1.59(0.4) = 0.636$ ounces
- Upper limit : $\bar{\bar{X}} + A_2 \bar{R} = 16 + 0.18(0.4) = 16.07$ ounces
- Lower limit : $\bar{\bar{X}} - A_2 \bar{R} = 16 - 0.18(0.4) = 15.93$ ounces

Out of control

Example

- A quality inspector took five samples, each with four observations, of the length of time to process a loan application. All values are in minutes. Use this information to obtain the three-sigma control limits for means of future times.

Sample 1	Sample 2	Sample 3	Sample 4	Sample 5
12.11	12.15	12.09	12.12	12.09
12.10	12.12	12.09	12.10	12.14
12.11	12.10	12.11	12.08	12.13
12.08	12.11	12.15	12.10	12.12

$$\begin{aligned}n &= 4 \\A_2 &= 0.729 \\D_3 &= 0 \\D_4 &= 2.282\end{aligned}$$

$$\begin{aligned}UCL_R &= 0.11 \\LCL_R &= 0 \\UCL_{\bar{X}} &= 12.11 + 0.729(0.05) = 12.14 \\LCL_{\bar{X}} &= 12.07\end{aligned}$$

Control Charts for Attribute data

- Data are classified according to different attributes (pass/fail; good/poor)
- Inspection of the units in the sample is performed on an attributes basis (measure is discrete)
- The 3σ upper and lower limits are:

$$\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

$$\text{Center Line} = \bar{p}$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

Example

- Jane Morgan operates a machine that makes bolts. Her company is implementing a self-charting program, and she wants to begin keeping track of the percent defectives at her operation. She knows that with this type of process, they expect about 4 percent defectives, plus or minus some chance variation. Jane wishes to initially construct a p-chart with three standard deviation control limits, and she has prepared ten daily samples of 100 bolts each.

Example

Sample Number	Percent Defectives
1	4
2	3
3	3
4	6
5	1
6	9
7	5
8	12
9	4
10	3

Solution

$$\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$$

- Upper limit: $4+3(1.9596) = 9.88 \%$
- Lower limit: $4-3(1.9596) = -1.88$ or 0

Out-of-control process

Sample Number	Percent Defectives
1	4
2	3
3	3
4	6
5	1
6	9
7	5
8	12
9	4
10	3

Example

■ TABLE 7.1

Data for Trial Control Limits, Example 7.1, Sample Size $n = 50$

Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$	Sample Number	Number of Nonconforming Cans, D_i	Sample Fraction Nonconforming, $\hat{p}_i$
1	12	0.24	17	10	0.20
2	15	0.30	18	5	0.10
3	8	0.16	19	13	0.26
4	10	0.20	20	11	0.22
5	4	0.08	21	20	0.40
6	7	0.14	22	18	0.36
7	16	0.32	23	24	0.48
8	9	0.18	24	15	0.30
9	14	0.28	25	9	0.18
10	10	0.20	26	12	0.24
11	5	0.10	27	7	0.14
12	6	0.12	28	13	0.26
13	17	0.34	29	9	0.18
14	12	0.24	30	6	0.12
15	22	0.44		347	$\bar{p} = 0.2313$
16	8	0.16			

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2313 + 3\sqrt{\frac{0.2313(0.7687)}{50}} = 0.4102$$

$$\text{Center Line} = \bar{p} = 0.2313$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.2313 - 3\sqrt{\frac{0.2313(0.7687)}{50}} = 0.0524$$

Example

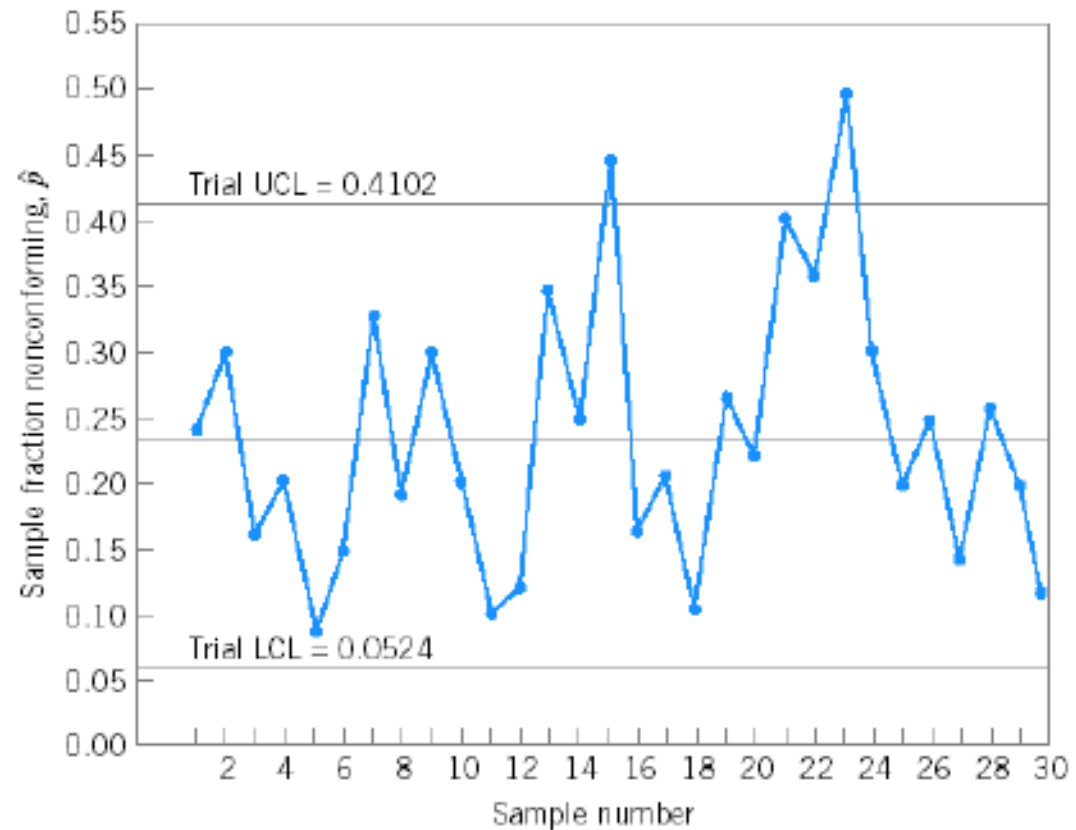
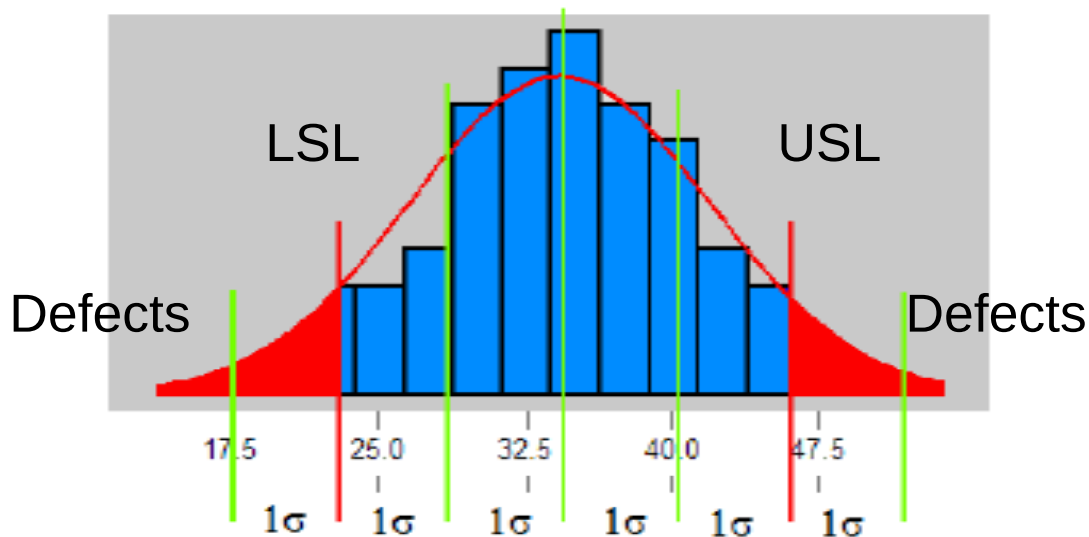


Figure 6-1 Initial phase I fraction nonconforming control chart for the data in Table 6-1.

- Not good at all, we must do something!

Process Capability

- Indicates how capable a process is in meeting customer requirements
- $C_p = \frac{USL - LSL}{6\hat{\sigma}}$
- USL: upper specification limit
- LSL: lower specification limit
- $\hat{\sigma}$: Process standard deviation



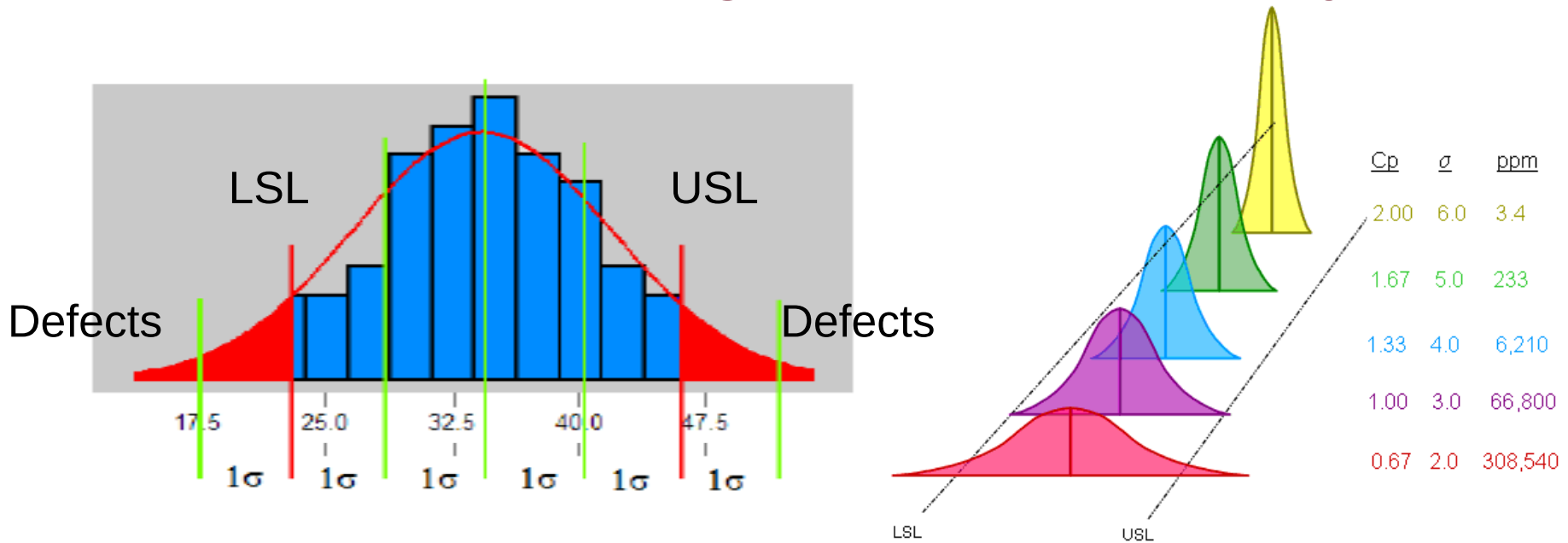
Six Sigma

- A quality initiative which aims for virtually error-free business performance (a standard of 3.4 problems per million)
- To make more money by improving effectiveness (customer value) and efficiency (the amount of resources consumed)
- “...*this Six Sigma journey will change the paradigm from fixing products so they are perfect to fixing processes that they produce nothing but perfection, or close to it.*” - Jack Welch (CEO- GE)

Six Sigma

- σ : Measures how far a process deviates from the mean, or from perfection
- Defect rate: no more than few parts (3.4) per million
- Achieve zero-defects
- Mean: production adjustment required
- Variance: system change required

Comparison between Six Sigma and traditional quality models

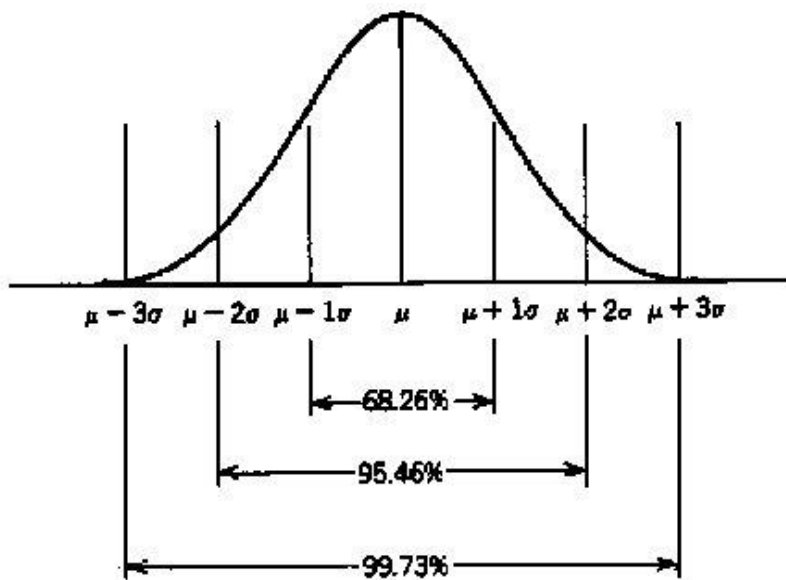


Tolerance=USL-LSL

- A way to measure the sigma level of a process is by calculating k in the formula below:
 - The distance from the target to the upper or lower specification (half the tolerance), measured in terms of standard deviation of the inherent variation

$$k^* \text{ process standard deviation} = \text{Tolerance} / 2$$

Comparison between Six Sigma and traditional quality models



—99% Good (3.8 Sigma)—

- 20,000 lost articles of mail per hour
- Unsafe drinking water for almost 15 minutes each day
- 5,000 incorrect surgical operations per week
- 340 Passengers with Misplaced Luggage every day
- 200,000 wrong drug prescriptions each year

—99.99966% Good (6 Sigma)—

- Seven articles lost per hour
- One unsafe minute every seven months
- 1.7 incorrect operations per week
- 6 Passengers with Misplaced luggage each month
- 68 wrong prescriptions per year

Six Sigma methodology

- Quality improvement via the DMAIC model
- A structured problem-solving process provides all employees with a common language and a set of tools to communicate with each other

